DATA 1030 Project

Team Any Idea: Building a Model to Predict New York City Energy Usage Intensity

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GitHub link

1. Introduction

New York City (NYC) has buildings serving residential, office, healthcare, retail, and other uses. Their energy demands vary depending on their uses and operating practices. When assessing energy use, building managers need to determine which properties require further investigation. However, energy intensity alone may not help building managers distinguish normal operational needs from potential problems without accounting for property characteristics. Therefore, our project aims to utilize NYC's building energy disclosure data for the 2024 calendar year to investigate whether property-level on-site energy use intensity (Site EUI) can be predicted based on property use, building and operational characteristics, and geographic location.

According to our research, previous studies have used building energy disclosure and property information to conduct energy forecasts. For example, Kontokosta and Tull (2017) analyzed electricity and natural gas usage by combining NYC's LL84 and PLUTO data, Ma and Cheng (2016) studied average residential Site EUI at the census block group level, and Robinson et al. (2017) trained models using data from the Commercial Buildings Energy Consumption Survey (CBECS) and validated them using NYC LL84 data. These studies support using property characteristics to predict energy use intensity. Our project aims to predict annual property-level Site EUI using NYC's 2024 LL84 dataset. We would find out how the property use, size, and other characteristics impact the predictions and evaluate the models' performance on other unseen data for better generalization.

2. Data

Our team used data published by the NYC Department of Buildings on the NYC Open Data platform. Property owners or their representatives submit annual benchmarking information through the EPA's ENERGY STAR Portfolio Manager. The data we used consists of records for the 2024 calendar year from NYC's LL84 Building Energy and Water Disclosure data. We retrieved the data via an API and filtered the data by the 2024 calendar year. In the data, each row represents a record for a reported property, and a property may consist of one or more buildings.

The original retrieved dataset contained 39,090 records and 256 columns. After cleaning, the dataset used for exploratory analysis consisted of 19,051 records and 44 columns, including one target variable, 39 numerical candidate predictor variables, and 4 categorical candidate predictor variables.

Our target variable is property-level Site Energy Use Intensity (Site EUI) for 2024. It is a continuous numerical variable calculated as annual site energy use divided by gross floor area,

measured in kBtu/ft². It shows energy use per unit of floor area rather than total energy consumption.

The candidate predictors cover property use, building and operational characteristics, and geographic location. For example, primary property type identifies a property's main use, gross floor area measures its size in square feet, and year built records its construction year. During data cleaning, we decided to retain existing properties that passed selected quality alert checks while removing records with missing values in the target variable, geographic information, or selected fields applicable to their property uses as the dataset is large enough. We also removed identification fields and redundant variables. To reduce the risk of target leakage, we excluded variables such as total site energy use, which is used to calculate Site EUI.

3. Exploratory Data Analysis

3.1 Dataset Overview

We conducted EDA on the cleaned dataset of 19,051 property records, focusing on Site EUI distribution, differences across property types, and the relationship between construction year and Site EUI to identify potential predictors and modeling challenges.

3.2 Target Variable Distribution

The median Site EUI is 68.2 kBtu/ft², while the mean is 128.13 kBtu/ft². The distribution is extremely right-skewed (skewness = 77.63), with a maximum value of 251,650 kBtu/ft².

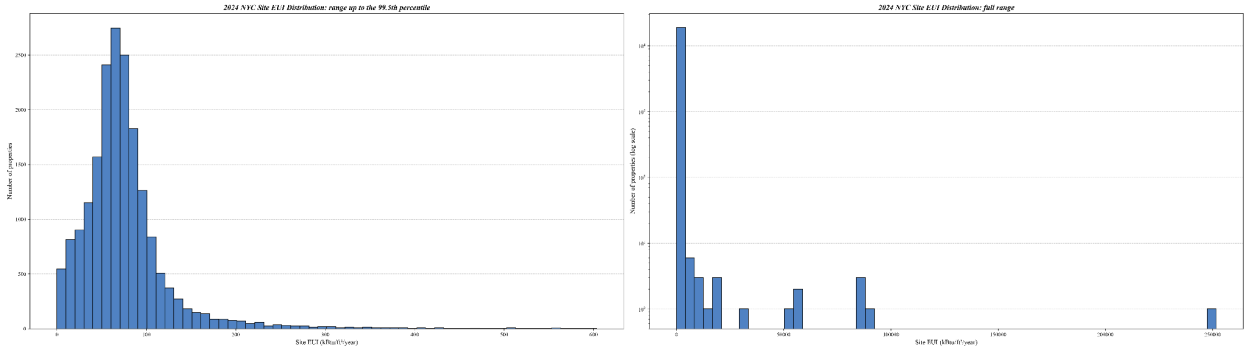


Figure 1. *Site EUI distribution up to the 99.5th percentile (left) and across the full range with a logarithmic y-axis (right).*

As shown in Figure 1, 99.5% of properties (18,955 out of 19,051) have Site EUI values at or below 603.45 kBtu/ft², while only 96 properties exceed this threshold. The left panel focuses on

these observations to better illustrate the distribution of typical values, while the right panel displays the full range to highlight the extreme right tail.

This finding is important for our prediction task because extreme values may disproportionately influence model training and evaluation, particularly when using squared-error metrics. These observations should therefore be investigated before determining an appropriate modeling and evaluation strategy.

3.3 Relationships Between Features and Site EUI

Property Type and Site EUI

We compared median Site EUI across property types with at least 100 observations.

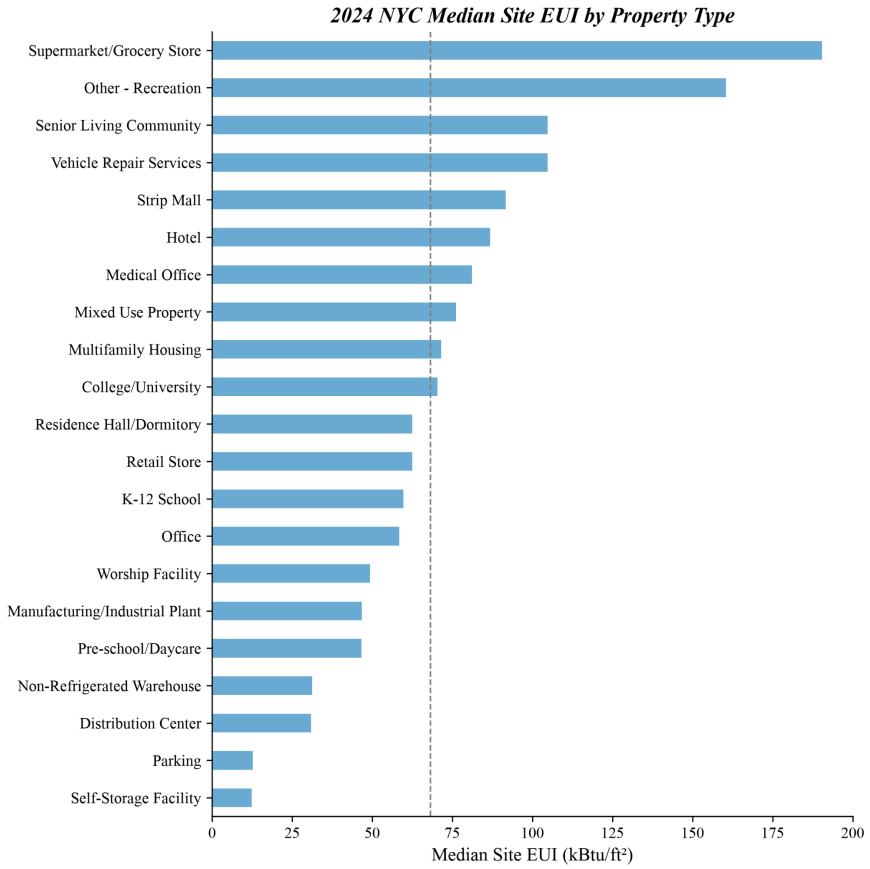


Figure 2. *Median Site EUI by property type (n ≥ 100 per type). The dashed line indicates the overall median of 68.2 kBtu/ft².*

Figure 2 shows substantial differences in median Site EUI across property types. Supermarket/Grocery Stores have the highest median Site EUI, approximately 190 kBtu/ft², followed by Other - Recreation properties at approximately 160 kBtu/ft². In contrast,

Self-Storage Facilities and Parking properties have the lowest median values, around 12 kBtu/ft². Multifamily Housing is slightly above the overall median, while Office buildings are below it.

This finding supports including property type as a predictor, since accounting for different building functions may help the model estimate expected Site EUI more accurately. Properties have different operating schedules, equipment requirements, and energy demands. For example, supermarkets may require continuous refrigeration, while storage facilities generally have fewer energy-intensive activities.

Construction Year and Site EUI

We also investigated the relationship between construction year and Site EUI among multifamily housing properties. By focusing on a single property type, we aimed to reduce variation caused by differences in building functions.

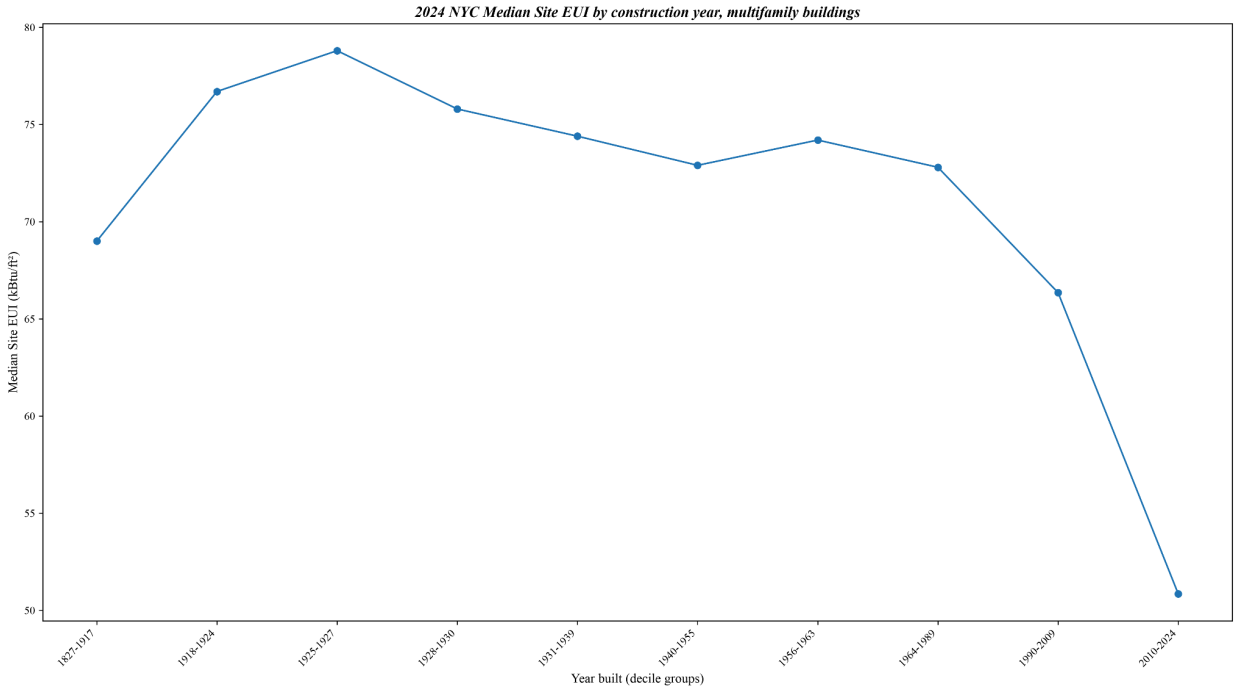


Figure 3. *Median Site EUI by construction-year group for multifamily housing properties.*

As shown in Figure 3, the relationship between construction year and median Site EUI is not strictly monotonic. Properties constructed between 1925 and 1927 have a relatively high median Site EUI of approximately 79 kBtu/ft², while those constructed between 2010 and 2024 have a substantially lower median of approximately 51 kBtu/ft².

Although the trend fluctuates across construction periods, the most recently constructed multifamily properties generally exhibit lower median energy use intensity than many older

groups. However, the relationship does not establish causality, as other factors such as renovations, heating systems, building design, and operational characteristics may influence energy consumption.

3.4 Key Findings

Our exploratory analysis provides three main findings.

First, Site EUI is extremely right-skewed, with a small number of exceptionally high values. This highlights the need to investigate extreme observations and carefully consider their influence on model training and evaluation.

Second, median Site EUI varies substantially across property types, suggesting that building function may be an informative predictor. Accounting for these differences may allow the model to produce more meaningful estimates for properties with different energy requirements.

Third, construction year appears to be associated with Site EUI among multifamily housing properties. Although newer properties tend to have lower median Site EUI in the most recent construction-year groups, the relationship is not strictly monotonic, suggesting that other characteristics may also play important roles.

Overall, these findings support the use of building characteristics to predict Site EUI while highlighting potential challenges related to extreme observations and differences across property types. They will help guide feature selection, preprocessing, and model evaluation in the next stage of our project.

4. Problem Description and Potential Impact

4.1 Supervised Learning Problem

Our team aims to develop a supervised machine learning regression model to predict the annual Site Energy Use Intensity (Site EUI) of existing properties in New York City. We will use records with reported Site EUI values to train a model and evaluate its predictions on properties not used for training. Because the target variable, site_eui_kbtu_ft, is continuous, this is a supervised regression problem. Operational inputs, such as occupancy and weekly operating hours, would need to reflect the year being assessed and be available when predictions are made. Any candidate features unavailable to the intended users would be excluded before modeling.

4.2 Potential Impact

One potential application is identifying properties whose actual Site EUI differs substantially from the value predicted based on their characteristics. For example, a property with an actual Site EUI considerably higher than its predicted value may warrant further investigation. Such discrepancies could indicate potential energy inefficiency, unusual operating conditions, or inaccurate reporting. However, a large discrepancy does not necessarily prove inefficiency, since the model may not capture every factor affecting energy demand.

The primary potential users include building managers and energy auditors. Building managers could use the predictions to compare the energy performance of properties they oversee against expected values for similar properties. Energy auditors could use these comparisons to prioritize properties for further assessment. Policymakers may also benefit from identifying broader patterns in building energy use across the city.

5. References

Kontokosta, C. E., & Tull, C. (2017). A data-driven predictive model of city-scale energy use in buildings. *Applied Energy*, *197*, 303–317. https://doi.org/10.1016/j.apenergy.2017.04.005

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